Algorithms Design and Analysis - HW IV

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Winter 2025-2026

Due to: 01/01/2026

Please submit the assignment in pairs, and make sure to include both of your names and ids on the PDF. 5 bonus points are given for a typed out solution, and my respect to whoever writes in LATEX. Make sure to prove each claim you use, unless it was seen in class. You may contact me for every question you have about the H.W.

Problem 1

Answer $True$ or $False$ for the following claims, and justify your answer with a proof or a counterexample:

1. In a graph with positive edge weights, any minimal-weight subset of edges that connects all vertices in the graph is a tree.

2. In a graph with arbitrary edge weights (which may include negative or zero weights), any minimal-weight subset of edges that connects all vertices in the graph is a tree.

Problem 2

Input: $G = (V, E)$ not directed graph. with positive values on the edges. The weight of the edges represent the distance it takes to drive the road e. In addition, we are given group $A \subset V$ of Football fields. Output: For each $v \in V$ return the closest football field (closest = the smallest distance we travel).

Problem 3

Let G be a positively weighted graph, and let T_1 and T_2 be two minimum spanning trees of G. Let $e \in T_1$ and $e \notin T_2$. Prove that there exists an edge $e' \in T_2$ such that $w(e) = w(e')$.

Problem 4

Let $G = (V, E, w)$ be a directed, non-negative weighted graph, which represents a system of roads and intersections. The weights of the edges define the travel cost along the corresponding roads, and let $s \in V$ be a given starting vertex. A subset of edges $F \subset E$ is given, representing roads defined as **congestion roads**.

- Additionally, the constants m and M are provided.

The travel cost in the road system for a path a is defined as follows:

- If a contains no congestion roads, the travel cost is the sum of the weights of the edges in the path.
- If a contains exactly one congestion road, the travel cost is the sum of the weights of the edges in the path plus m.
- If a contains at least two congestion roads, the travel cost is the sum of the weights of the edges in the path plus M.

Propose an efficient algorithm to compute the minimum travel cost from s to every other vertex in the graph.

Problem 5

Given a Directed Acyclic Graph $G = (V, E)$ and a vertex s, the goal is to compute the shortest (with respect to edge weights) path from s to all vertices in the graph.
Note that the Bellman-Ford algorithm solves this problem in $O(V \cdot E)$. However, you should find an algorithm with a lower time complexity.